

Egypt Real Estate Listings

Data Mining Course — 2024 / 2025

Project Team

Task	Member(s)
Task 1: EDA & 5 Visualizations	Abdelrahman Mohamed
Task 2: Data Cleaning & Preprocessing	Abdelrahman Mohamed
Task 3: K-Medoid Clustering	Youssef Salah
Task 4: Hierarchical Clustering	Abdallah Mohamed Elsiisy Youssef Anwar
Task 5: Fuzzy Logic System	Ahmed Khamis Ahmed Yasser
Task 6: Genetic Algorithm	Khaled Hafez Assem Mohamed Gamal
Task 7: System Implementation & Pipeline	Mohamed Rizk Youssef Shaban

1. Introduction

1.1 Dataset

The dataset is the Egypt Real Estate Listings dataset sourced from PropertyFinder.eg and published on Kaggle by Hassan Khaled. It contains 19,924 property listings covering apartments, villas, chalets, and other property types across multiple Egyptian governorates.

Attribute	Value
Source	PropertyFinder.eg (via Kaggle — Hassan Khaled)
Instances	19,924 listings
Raw Features	11
Features After Preprocessing	~30 engineered features
Domain	Real Estate / Investment Analysis

1.2 Raw Features

Feature	Type	Description
price	Numeric (string)	Listing price in EGP — formatted with commas
type	Categorical	Property category: Apartment, Villa, Chalet, etc.
location	Categorical	Full address — compound, district, city, governorate
size	Numeric (string)	Area in sqft and sqm (both in one string)
bedrooms	Numeric	Bedroom count — may include maid room (e.g. '3+Maid')
bathrooms	Numeric	Bathroom count
payment_method	Categorical	Cash or Installments
down_payment	Numeric	Down payment in EGP — null for cash listings
available_from	Date	Delivery or availability date
url & description	Text	Dropped — no modeling value

1.3 Domain Motivation

The Egyptian real estate market has experienced significant price fluctuations due to inflation. Dividing properties into meaningful categories and developing a valuation plan is crucial for both investors and buyers.

2. Exploratory Data Analysis (Task 1)

Before modeling, five visualizations were produced to understand the data's structure and distributions. Temporary numeric conversions were used for plotting only — permanent transformations were done later in preprocessing.

Visualization 1 — Price Distribution

A histogram of property prices showed a heavily right-skewed distribution. Most listings are priced below 20M EGP, with a small number of ultra-luxury properties forming a long right tail. The median is used as the central measure (more reliable than mean for skewed data). This skew is expected in real estate and directly justified the use of a 3.0 IQR multiplier (instead of the standard 1.5) for outlier removal.

Visualization 2 — Property Type Frequency

A bar chart showed that Apartments dominate the market, followed by Chalets and Villas. This reflects Egypt's urban housing structure where dense residential apartments far outnumber vacation or luxury types. The wide variety of property types confirms that the dataset covers a broad market spectrum — an important factor when building clusters.

Visualization 3 — Size vs. Price Scatter

A scatter plot with Pearson r value showed a moderate positive correlation between property size and price — larger properties generally cost more. However, the wide spread at larger sizes confirms that size alone does not fully explain price. Location, property type, and finishing quality are also significant drivers.

Visualization 4 — Payment Method Breakdown

A pie chart revealed that around 80% of listings are cash-based. This shows that the Egyptian real estate market is predominantly cash-driven, and also explains why most `down_payment` values are null — they only apply to installment listings. This insight shaped how missing `down_payment` values were handled (filled with 0).

Visualization 5 — Top 10 Locations

A horizontal bar chart showed that New Cairo, North Coast, and Sheikh Zayed are the most active listing areas — Egypt's primary premium real estate corridors. New Cairo's dominance reflects rapid urban development east of Cairo. The North Coast's strong presence indicates high demand for vacation and second-home properties. This guided the governorate encoding strategy.

3. Data Preprocessing (Task 2)

Raw data from PropertyFinder.eg was heavily unstructured and required multiple cleaning and transformation steps before any modeling could be applied.

3.1 Cleaning Steps

Step	What Was Done	Why
Drop irrelevant columns	Dropped url and description	No modeling value
Fill down_payment nulls	Filled with 0	Null means cash listing — no down payment
Fill available_from nulls	Filled with mode value	Avoids data loss for a non-critical column
Drop critical nulls	Dropped rows with null price, location, type, size, bedrooms, bathrooms, payment_method	Core features — imputation would add noise
Remove duplicates	Dropped exact duplicate rows	Prevent bias in model training

3.2 Transformations

Feature	Before	After	Why
price	'2,500,000 EGP'	2500000.0 (float)	Cannot do math on formatted string
size	'150 sqft / 120 sqm'	120.0 sqm	sqft is redundant — sqm is consistent
bedrooms	'3+Maid'	3 (int) + has_maid=1	Extract number; maid flag is useful feature
available_from	'2025-06-01'	delivery_month=6, delivery_year=2025	Date decomposed into usable numerics
payment_method	'Cash' / 'Installments'	0 / 1 (label encoded)	Binary variable — label encoding is sufficient
type	Apartment, Villa...	One-hot encoded columns (type_...)	Nominal — no ordinal relationship
governorate	Extracted from location string	One-hot encoded; rare (<100 listings) grouped into Other	Reduces noise from rare categories

3.3 Outlier Removal

IQR-based outlier removal was applied to price and size using a 3.0 IQR multiplier — deliberately more conservative than the standard 1.5. Real estate data is naturally right-skewed due to luxury properties. Using 1.5 IQR would aggressively remove valid premium properties. The 3.0 multiplier filters only genuine data entry errors while preserving the legitimate luxury market segment.

3.4 Feature Scaling

StandardScaler was applied only to the 7 continuous numeric columns (price, size, bedrooms, bathrooms, down_payment, delivery_year, delivery_month). The binary dummy columns (0/1) were kept unchanged and concatenated after scaling. Reason: K-Medoids is distance-based — without scaling, price (in millions) would completely dominate the distance calculation and overshadow all other features.

3.5 Dimensionality Reduction (Bonus)

An unscaled copy of the cleaned data (df_clean_unscaled) was saved before scaling for use in the Fuzzy Logic system. The Fuzzy membership functions require human-readable values (actual EGP prices, actual sqm), not z-scores.

4. Clustering

4.1 K-Medoids Clustering (Task 3)

Why K-Medoids?

K-Medoids was chosen over K-Means for two reasons. First, K-Means uses mathematical means as centroids — for binary dummy columns (like governorate one-hot features), a centroid of 0.3 is meaningless. K-Medoids uses actual data points as cluster centers, making results interpretable. Second, K-Medoids is more robust to outliers since it minimizes total absolute distances rather than squared distances.

Choosing K — Elbow Method + Silhouette Score

Both methods were applied across K = 2 to 10. The global Silhouette maximum was at K=2, but this produces an oversimplified binary split (cheap vs. expensive) with no actionable market insight. K=5 was selected based on a local Silhouette maximum (score = 0.1045) and a meaningful flattening of the Elbow inertia curve. K=5 provides five distinct market tiers that reflect real segments in the Egyptian property market.

Cluster Profiles (K=5, Manhattan distance)

Cluster	Likely Segment	Key Characteristics
Cluster 0	Affordable Residential	Low price, small-medium size, few bedrooms
Cluster 1	Mid-Market Apartments	Mid-range price, standard size, Cairo-dominated
Cluster 2	Premium / Luxury	High price, large size, many bedrooms
Cluster 3	Vacation / Chalet	North Coast-heavy, mid-price, smaller size
Cluster 4	Suburban Villas	Large size, good value per sqm, Giza-area

4.2 Hierarchical Clustering (Task 4)

Three linkage methods were applied on a 2,000-sample subset (full dataset is too large for linkage computation). Dendrograms were plotted for each. The same K=5 was used for the cut, consistent with the K-Medoids decision.

Linkage Method	Behavior	Result at K=5
Ward	Minimizes within-cluster variance at each merge	Most compact, balanced clusters — best result
Complete	Uses maximum distance between clusters	Sensitive to outliers, less balanced
Average	Uses mean distance between all pairs	More chain-like clusters, lower cohesion

The Ward linkage dendrogram confirmed the K=5 cut is natural — the jump in merge distances at that point was clearly visible. Hierarchical clustering validated the same broad market structure identified by K-Medoids.

5. Fuzzy Logic Inference System (Task 5)

A Mamdani Fuzzy Inference System was designed to assess the investment quality of any given property. It uses three inputs from `df_clean_unscaled` (unscaled, human-readable values) and outputs a score on a 0–10 scale.

5.1 System Overview

Component	Value
Inference Type	Mamdani
Defuzzification	Centroid method
Inputs	3 — price, size, bedrooms
Rules	8
Output	Investment_Opportunity (0–10 scale)

5.2 Membership Functions

Variable	Type	Terms	Method
price (EGP)	Input	Low / Medium / High	Trapezoid at P33 and P66 percentiles
size (sqm)	Input	Small / Medium / Large	Trapezoid at P33 and P66 percentiles
bedrooms	Input	Few / Average / Many	Triangular — Few: 1–3, Average: 2–5, Many: 4+
Investment_Opportunity	Output	Bad / Average / Excellent	Trapezoid — Bad: 0–4, Average: 3–7, Excellent: 6–10

5.3 Rule Base

The 8 rules encode intuitive real-estate investment logic grounded in domain knowledge:

Rule	IF Condition	THEN Output
R1	price = Low AND size = Large	Excellent — cheap + big = great deal
R2	price = Low AND size = Medium	Excellent — cheap + decent size = good value
R3	price = Low AND beds = Many	Excellent — cheap + many rooms = great deal
R4	price = High AND size = Small	Bad — expensive + tiny = poor value
R5	price = High AND beds = Few	Bad — expensive + few rooms = poor value
R6	price = Medium AND size = Medium	Average — mid-range in both = average deal
R7	price = Medium AND beds = Average	Average — balanced mid-range property
R8	price = High AND size = Large	Average — expensive but big = fair value

5.4 Validation

The system was tested on a random sample of 500 properties from `df_clean_unscaled`. Each property's price, size, and bedrooms were fed into the fuzzy system and a score was computed using centroid defuzzification. Results showed a meaningful spread across Bad / Average / Excellent categories, consistent with the rules.

Properties with high price and small size scored Bad; low-price large properties scored Excellent. A try/except fallback (score = 5.0) was added to handle edge cases where inputs fall in a region not covered by any rule.

6. Genetic Algorithm — Feature Selection (Task 6)

A Genetic Algorithm was applied to perform feature selection on the combined dataset (df_combined). The goal is to find the optimal subset of features that maximizes the Silhouette Score of K-Medoids clustering. Using all features introduces noise — some governorate dummies or property type flags may degrade cluster quality. The GA searches the exponential feature space efficiently to find the best subset.

6.1 GA Design

Component	Value	Rationale
Chromosome	Binary array — length = number of features	1 = include feature, 0 = exclude
Gene	One bit per feature	Represents inclusion/exclusion of that feature
Population Size	50 chromosomes	Sufficient for the feature space size
Generations	20	Enough to show clear convergence
Fitness Function	Silhouette Score of K-Medoids (K=5)	Direct measure of cluster quality
Selection	Tournament (k=3)	Balances selection pressure and diversity
Crossover	Single-point (rate = 0.8)	Standard — preserves partial good solutions
Mutation	Bit-flip per gene (rate = 0.08)	Introduces diversity to avoid local optima
Children	Top 2 kept (elitism) + offspring fill rest	Prevents losing best solution found
Minimum Features	5 features required	Penalizes degenerate near-empty chromosomes

6.2 Results

Metric	Value
Baseline Silhouette (all features)	Computed before GA run
Best GA Silhouette Score	Best score across all 20 generations
Improvement over baseline	Positive — GA found a better feature subset
Convergence	Visible upward trend — plateau reached by generation ~15–18

The GA convergence plot showed a clear improvement from generation 1 to the plateau. The feature selection bar chart identified which features (price, size, bedrooms, and a subset of categorical dummies) contributed most to cluster quality.

7. System Implementation — Pipeline (Task 7)

The `run_real_estate_pipeline()` function integrates all four components into one end-to-end workflow. It takes a raw property record and produces a market cluster assignment and an investment recommendation.

7.1 Pipeline Design — Dual-Pathway Architecture

The pipeline uses two parallel pathways that work independently and then merge into a final report:

Pathway	Steps	Purpose
Expert Logic Path (Fuzzy)	Raw values → Fuzzy System → Investment Score	Uses human-readable values. Gives an investment rating based on domain rules.
Statistical Logic Path (ML)	Scale → Encode → GA Filter → K-Medoids → Cluster	Uses scaled/encoded features. Assigns the property to one of 5 market segments.

7.2 Pipeline Steps

Step	Description
1 — Input	Receive raw property attributes: price, size, bedrooms, bathrooms, down_payment, delivery_year, delivery_month, type, governorate.
2 — Fuzzy Scoring	Feed price, size, bedrooms directly into the Fuzzy System (unscaled). Compute Investment_Opportunity score (0–10).
3 — Scale + Encode	Apply the fitted StandardScaler on numeric features. One-hot encode categorical features. Align columns to training schema.
4 — GA Filter	Apply the best_chrom GA mask — keep only the selected feature indices.
5 — Cluster Assignment	Run <code>kmed_ga.predict()</code> to assign the property to one of 5 market segments.
6 — Final Report	Print asking price, size, market cluster, fuzzy score (0–10), and recommendation (Bad / Average / Excellent).

7.3 Flowchart

```
INPUT: price · size · bedrooms · bathrooms · down_payment · type · governorate
↓
PATHWAY 1 (Expert Logic) PATHWAY 2 (Statistical)
Fuzzy System Scale + One-Hot Encode
(price, size, bedrooms unscaled) Apply GA Feature Mask
→ Investment Score (0–10) → K-Medoids Predict
↓ ↓
MERGE: Market Segment + Investment Score
↓
OUTPUT: Cluster [0-4] | Score X.XX/10 | Recommendation
```

7.4 Sample Test

A test property was run through the pipeline:

Input Field	Value
price	4,200,000 EGP

Input Field	Value
size	160 sqm
bedrooms	3
bathrooms	2
down_payment	420,000 EGP
type	Apartment
governorate	Cairo
delivery	June 2025

The pipeline outputs: Market Cluster assignment, Fuzzy Score out of 10, and a recommendation label (Bad / Average / Excellent).

8. Conclusion

Section	Key Finding
EDA (Task 1)	Price is right-skewed; apartments dominate; New Cairo, North Coast, and Sheikh Zayed are the most active markets.
Preprocessing (Task 2)	Raw data required heavy parsing (strings to floats, compound strings to extracted values). A 3.0 IQR multiplier preserved the luxury segment.
K-Medoids (Task 3)	K=5 was selected using a local Silhouette maximum — balancing mathematical validity with domain-meaningful segmentation.
Hierarchical (Task 4)	Ward linkage produced the most compact clusters. All three methods confirmed the K=5 structure. The dendrogram validated the cut point.
Fuzzy Logic (Task 5)	8 domain-grounded rules effectively differentiate investment quality. Low-price large properties score Excellent; high-price small properties score Bad.
GA (Task 6)	Feature selection improved Silhouette Score over the all-features baseline by filtering out noisy governorate dummies.
Pipeline (Task 7)	A dual-pathway system merges fuzzy scoring (unscaled) and ML clustering (scaled + GA-filtered) into a unified investment report.

Business Value

- Investors can screen any property in seconds — getting both a cluster segment and an investment score.
- The fuzzy system flags bad deals (e.g. high price, small size) even within premium clusters.
- The 5-cluster segmentation allows targeted marketing and pricing analysis per market tier.
- The GA-selected features improve clustering robustness by removing noise from rare location dummies.

End of Report